

Part 5 Answers

1. What is the slope?

The slope is represented by the coefficients of the Linear Regression model. It shows how much the predicted marks change when a feature increases by one unit while keeping other features constant.

Slope Values:

Study_Hours = 3.45487968

Attendance = 1.34491979

Assignments = -2.40274064

2. What is the intercept?

The intercept is the predicted value of marks when all feature values are zero.

Intercept Value:

-48.100601604278054

3. What do they mean?

The coefficients indicate the influence of each feature on marks. Positive coefficients increase predicted marks, while negative coefficients decrease them. The intercept serves as the baseline prediction.

Part 7 Answer

Which student had the highest prediction error?

The student with Actual Marks = 35 had the highest prediction error.

Actual Marks = 35

Predicted Marks = 31.243984

Difference = 3.756016

Part 8 Answers

1. What is MAE?

Mean Absolute Error (MAE) measures the average absolute difference between actual and predicted values. Lower values indicate better accuracy.

2. What is R^2 Score?

R^2 Score measures how well the model explains the variation in the target variable. Values closer to 1 indicate better performance.

3. Is your model performing well?

Yes. The model performs very well because it achieved a low MAE and a high R^2 Score.

MAE = 1.3453654188948245

R^2 Score = 0.9929962957709836